

Evaluating Compressed Neural Networks Under Measured Energy and Reliability Constraints

Toward Deployment-Aware Model Selection: A Controlled, Hardware-Measured Study of Energy and Calibration Trade-offs in Compressed Vision Models

Master's Research Proposal — Two-Year Programme

Shohinur Pervez Shohan

B.Sc. in Computer Science & Engineering, Rangamati Science and Technology University,
Bangladesh

Prospective Applicant, 2027 Intake

Abstract

Making a neural network smaller can reduce the computing power it needs — but it can also change how confidently the model is wrong. This proposal tests whether compressed image-classification models that achieve similar accuracy and similar measured energy consumption remain equally trustworthy once their confidence, not just their correctness, is examined under realistic image degradation such as blur, noise, and compression artefacts. This matters directly for anyone deploying vision models on energy- or battery-constrained devices: an energy saving is only useful if it does not conceal more confident mistakes.

A three-seed pilot study, completed on the applicant's own hardware ahead of this application, already shows a small but directionally consistent signal in exactly this direction — compressed models miscalibrated relative to a dense baseline, contradicting the specific direction reported by the closest prior study. This proposal sets out the confirmatory design that follows from that pilot: a controlled comparison, at matched measured energy, of what compression actually buys and what it may quietly cost.

Phase 1 (months 1–12) is the committed thesis deliverable and extends work already under way on hardware the applicant owns. Phase 2 (months 13–24) is explicitly conditional on supervision and resources the applicant does not independently possess, and is scoped to one selected extension rather than a broad programme.

1. Introduction

Deep neural networks have become both more capable and more expensive. Deployment on edge devices, mobile platforms, and energy-constrained infrastructure depends on compression: pruning removes parameters, quantization reduces numerical precision, and knowledge distillation transfers behaviour from a large teacher to a compact student.

Evaluation of these methods has matured along two separate tracks. One community measures efficiency — FLOPs, parameter count, latency, and increasingly energy on real hardware. The other measures reliability — calibration, robustness under distribution shift, and class-level degradation. Both communities have produced substantial results. Neither

has produced results that answer the question a practitioner actually faces at deployment time.

The practical question is this: *when two compression configurations achieve comparable accuracy at comparable measured energy cost, is one of them a worse choice once the trustworthiness of its confidence is examined?* Answering it requires measuring both properties on the same models, under one protocol, with an energy figure whose limits are understood rather than assumed. This proposal reports preliminary evidence that the answer may be yes, and sets out the confirmatory design that follows.

2. Preliminary Work and Evidence to Date

This section reports work already completed on the applicant’s own hardware, prior to submission. It exists because an earlier version of this proposal made a claim about completed pilot work without supplying evidence — a fault identified in review and corrected here by reporting only what was actually run, with real output values, on real data.

2.1 Hardware and environment, as verified

Component	Verified specification
CPU	Intel Core i5-13450HX (10C/16T)
GPU	NVIDIA RTX 3050 6GB Laptop GPU, driver 595.84
RAM	24GB DDR5
OS	Linux Mint 22.3 (Ubuntu 24.04-based), kernel 7.0.0-31-generic, bare metal (confirmed via systemd-detect-virt)
RAPL access	3 zones (package-0, core, psys), world-readable, confirmed non-root; udev rule installed
NVML access	nvmlDeviceGetTotalEnergyConsumption confirmed functional against real workloads
Software	PyTorch 2.7.1+cu118, torchvision, onnxruntime-gpu 1.29.0

Both target datasets (CIFAR-10, 170.5MB; CIFAR-10-C, 2.92GB, all 15 corruption types at 5 severities) are downloaded, checksum-verified, and extracted on this machine.

2.2 Execution feasibility: tested, not inferred

Prior to committing to an experimental grid, every proposed compression configuration was tested directly on this hardware rather than assumed from documentation. This surfaced three findings that materially changed the design, each verified through dtype inspection or profiler traces rather than timing alone.

Configuration	Status	Evidence
FP32 / FP16 (GPU, both architectures)	Executable	Verified via weight/activation dtype inspection, not autocast simulation. ResNet-18: 1.71× real speedup; MobileNetV3-Small: 1.75×
INT8, ResNet-18 (CPU only, fbgemm)	Executable	Real quantized::conv2d / quantized::linear ops confirmed via torch.profiler trace, not QDQ-around-FP32. 1.39× speedup
INT8, MobileNetV3-Small	Dropped from committed scope	No quantization-ready variant exists in torchvision (only mobilenet_v3_large has one); would require custom model surgery
INT8, GPU (either architecture)	Dropped — confirmed dead end	fbgemm-quantized model segfaults on .to(“cuda”) (exit 139). ONNX Runtime + TensorRT: CUDA/TensorRT execution providers silently fall back to CPU despite being listed as available, due to a CUDA 11.8/13 mismatch
INT4 (any architecture, any device)	Dropped — confirmed dead end	No Conv2d INT4 kernel exists in any installed or installable mainstream library (torchao, bitsandbytes, auto-gptq); bitsandbytes only implements Linear4bit / Embedding4bit
Native PyTorch pruning API (structured or unstructured)	Executable but confirmed to give zero benefit	torch.nn.utils.prune masks weights with zeros that remain densely stored and computed. 70% unstructured sparsity: 0.987× latency ratio (statistical noise). The “structured” variant only masks channels without resizing tensors — identical null result
Real structural (channel) pruning, via torch-pruning	Executable, adopted	Dependency-graph-aware channel removal, genuine tensor shrinkage confirmed. ResNet-18 at 30/50/70% target ratios: real speedups of 0.83×, 0.43×, 0.29× baseline latency (i.e. up to ≈ 3.5× faster at 70%). MobileNetV3-Small: 0.74×, 0.53×, 0.32×

This is treated as a finding, not merely a scoping exercise. The compression method used in the closest prior calibration study (§2.3) — unstructured, global-magnitude weight pruning — is exactly the method confirmed above to yield zero measured energy or latency benefit on consumer hardware, regardless of the nominal sparsity ratio reported. A “30% pruned” model can therefore mean two operationally unrelated things: a model with 30% of its weights zeroed but unchanged compute cost, or a model with 30% of its channels physically removed and genuinely cheaper to run. This distinction, confirmed empirically rather than assumed, motivates separating the reliability question (§2.3, using the unstructured method to enable direct comparison with prior work) from the energy-linked question (§6, WP3, using confirmed real structural pruning) rather than conflating them under one label.

2.3 Reconstruction of the closest prior study

Mitra, Schwalbe & Klein (2024) is the closest existing work to this thesis (§4.2). Their reported protocol — VGG-19-BN, CIFAR-10, global unstructured L1 conv-weight pruning at 30%, 160 baseline epochs followed by 40 recovery epochs — was independently reconstructed and run to completion for three seeds, with full clean and CIFAR-10-C (15 corruptions × 5 severities) evaluation. This is not the authors’ released code; it is an independent implementation built from their paper’s stated methodology, with full source and data provenance (SHA-256 hashes, commit references) recorded for every run.

Metric	Baseline (mean ± SD, 3 seeds)	Pruned 30% + recovered (mean ± SD, 3 seeds)	Direction across seeds
Clean accuracy	0.9372 ± 0.0010	0.9375 ± 0.0011	Inconsistent (2 of 3 seeds positive) — no detectable effect
Clean ECE (equal-mass)	0.0472 ± 0.0012 (across seeds)	—	Consistent: all 3 seeds show pruned ECE higher
Corruption mPC (mean over 5 severities)	Ranges 0.8678 – 0.5655 by severity	Comparable, within noise	Inconsistent at severities 1, 2, 5; consistent (worse) only at 3, 4
Pooled corruption ECE, all 5 severities	Lower at every severity	Higher at every severity	Consistent across all 3 seeds, at every severity — 6 of 6 conditions

The calibration direction is the result that survives replication. Across the clean condition and all five corruption severities — six independent comparisons — every one shows the pruned-and-recovered model less well calibrated than the dense baseline, in every one of three seeds. Under a null hypothesis of no systematic direction, six independent same-direction outcomes has probability $(\frac{1}{2})^6 \approx 0.016$. No individual per-severity paired t-test ($df = 2$) clears the conventional critical value at this seed count, and the magnitudes are small (absolute ECE differences of 0.0004–0.013); this is reported as suggestive directional evidence from a 3-seed pilot, not a confirmed effect, and is precisely why a 10-seed confirmatory design is proposed (§6) rather than presented as already established.

This directly contradicts the direction Mitra et al. report — that post-hoc pruning can improve calibration and corruption robustness simultaneously. One plausible, testable explanation for the discrepancy is protocol-level: in Mitra et al.’s design, the pruned model receives 40 additional epochs of recovery training that the baseline, as described, does not receive. If a dense model given the same 40 additional epochs shows comparable calibration change, the reported improvement may be partly attributable to extra training rather than to pruning itself — the same class of finding as the applicant’s undergraduate thesis, where a validation-design choice, not the algorithm, determined the reported

conclusion (§10). This is the central confirmatory hypothesis of Phase 1 (§6, H2), not yet resolved, and is proposed precisely because the preliminary evidence above is directional but underpowered.

2.4 Instrument characterisation

Two measurement-validity findings emerged directly from running the harness, independent of the substantive compression question, and are reported as preliminary methods results.

NVML short-window artifact. Measuring the identical idle GPU state at increasing polling intervals produced implied average power that should be constant but was not:

Polling interval	20ms	50ms	100ms	200ms	300ms	500ms
Implied avg. power	291.25W	126.72W	84.84W	51.22W	26.13W	15.84W

This GPU has a 60W cap, so the 20ms reading is physically impossible; a correctly behaving monotonic counter should telescope to the same total regardless of polling frequency. This is diagnostic of a genuine NVML/driver quantisation artifact on this consumer GPU, distinct from datacentre-class GPUs with dedicated fast power sensors. RAPL, tested identically, showed no equivalent artifact (1ms hardware update interval). The practical consequence, adopted throughout: no NVML-derived energy measurement is trusted below a 1-second measurement window.

GPU-contention diagnostic. An accidental concurrent-training incident during timing measurement was diagnosed rather than discarded: SM clock ceiling, not power draw, distinguishes contended from clean runs on this hardware, because both regimes saturate the 60W power cap identically. Clean runs reach 1792–1972MHz; a contended run was capped at 1627–1845MHz. This diagnostic was used to identify and then correct one contaminated measurement (§2.5).

2.5 Real energy pilot and validated compute budget

A corrected pilot run (0.5s polling interval, 2.0s measurement windows, 3 repeats) produced physically plausible, real values: idle draw 17–29W, active draw 71–78W, gross energy 0.0447 J/image, with 68% of total measured energy above the idle baseline. The harness correctly reports this pilot as not yet confirmatory (n = 3 paired measurements, below the pre-specified minimum of 30) — it establishes that the harness produces plausible numbers, not that any compression effect is yet resolved.

Per-epoch training time was measured directly, with contention explicitly ruled out via the GPU-clock diagnostic above:

Architecture	Per-epoch (measured, clean)	1 seed (200 epochs)	10 seeds, sequential
VGG-19-BN (reconciliation track, §2.3)	68.68s	3.82h	38.2h (1.59 days)
ResNet-18 (thesis architecture)	47.71s (SD 0.082s)	2.65h	26.5h (1.10 days)
MobileNetV3-Small (thesis architecture)	7.08s (SD 0.16s)	0.39h	3.9h (0.16 days)

Total sequential GPU time for the thesis’s own confirmatory training grid (ResNet-18 + MobileNetV3-Small, 10 seeds each): approximately 30.4 hours $\approx$ 1.3 days, unattended, measured directly rather than estimated — comfortably within a normal thesis schedule, with real margin for reruns.

3. Research Motivation

3.1 The measurement problem

Compression is routinely justified by reductions in FLOPs or parameter count. Direct hardware measurement has repeatedly shown these proxies unreliable: energy depends on memory traffic, kernel availability, cache behaviour, and batch size in ways operation counts do not capture (§4.3). This proposal takes that finding as an established premise, not a contribution.

The consequence has not been fully followed through. Software estimators such as CodeCarbon and pyJoules have been validated against physical meters on general AI workloads, with documented errors up to 40% (Fischer, 2025). They have not been tested on compressed models, where quantization alters the executed instruction mix. It is also necessary to be precise about what such a comparison can establish: pyJoules and, in part, CodeCarbon derive their readings from the same RAPL and NVML interfaces this proposal also reads directly. Comparing these tools against RAPL/NVML therefore audits accounting and sampling consistency between layers of the same measurement stack — a legitimate and, per §4.4, unaddressed question — rather than constituting independent physical validation. Independent validation against an external reference (a wall-plug or bench meter, with its measurement boundary explicitly reconciled against the GPU/CPU package boundary) is treated in this proposal as Phase 2 work (§7, WP5), not claimed in Phase 1.

3.2 The reliability problem

A compressed model that retains aggregate accuracy is conventionally treated as successful. Compression has been shown to cause selective forgetting concentrated in a small subset of classes (Hooker et al., 2019), to produce disparate accuracy impacts across groups (Tran, Fioretto & Kim, 2022), and to alter predictive behaviour in ways aggregate metrics do not surface.

The direction and magnitude of these effects is not settled by isolated comparison. Post-hoc pruning has been reported to improve calibration and corruption robustness in one study (Mitra, Schwalbe & Klein, 2024) and in embedded-inference settings (Huber, Göhner & Trapp, 2025). Other work reports reduced robustness under domain shift and adversarial perturbation, with quantization outperforming pruning (Shen et al., 2024). Still other work finds compression preserving or improving corruption robustness under different conditions (Diffenderfer et al., 2021; da Silva et al., 2025). These are context-dependent effects — obtained under different training procedures, compression intensities, and forms of distribution shift — rather than a strict contradiction resolvable by any single additional study; this proposal’s own reconstruction (§2.3) adds one more controlled data point in this space, under a protocol close enough to Mitra et al.’s to be directly informative about that specific comparison.

4. Related Work

This section states what prior work established and what it leaves open. Characterisations such as “not identified in our search” reflect systematic searches across Consensus, Scite, Scholar Gateway, Elicit, ResearchRabbit, and direct publisher retrieval, covering 2019–2026; the absence of a result in a search is not treated as proof of its absence in the literature.

4.1 Neural network compression evaluation

Compression evaluation is mature and extensively benchmarked, typically on accuracy, parameter count, FLOPs, and increasingly latency; joint pruning–quantization optimisation via reinforcement learning has achieved 39% average energy reduction for 1.7% average accuracy loss on embedded accelerators (Balaskas et al., 2023). Evaluation remains accuracy-centric, and reliability properties are rarely part of the comparison protocol. This work does not propose a new compression algorithm; it treats the evaluation protocol as the object of study.

4.2 Reliability consequences of compression

Hooker et al. (2019) showed pruning disproportionately degrades a small subset of classes while aggregate accuracy moves little — 170 ImageNet classes significantly affected at 50% sparsity, rising to 372 at 70%. Related work characterised bias through compression-identified exemplars (Hooker et al., 2020), class/cohort-level degradation (Paganini, 2020), and gradient-norm-mediated group disparity (Tran, Fioretto & Kim, 2022). None measures calibration or energy.

Mitra, Schwalbe & Klein (2024) is the closest existing work to this thesis, evaluating pruning of perception CNNs on expected calibration error and corruption robustness — reporting aggregate top-label ECE, not disaggregated per class, and not measuring energy. Huber, Göhner & Trapp (2025) measures embedded-system response time alongside ECE in a different application domain, without a multi-family compression sweep or energy measurement. Yuan et al. (2023) examined PTQ reliability with attention to calibration-set distribution and worst-case subgroup performance; Williams & Aletras (2023) established that calibration-data selection substantially affects compressed-model task performance —

a distinct concept from probability calibration, and a confound this proposal's own reconstruction (§2.3) is designed to isolate rather than assume away. Kamal & Talbert (2025) introduced faithfulness metrics beyond accuracy for compressed models on socially meaningful datasets, without energy measurement.

4.3 Energy measurement and hardware-aware evaluation

Tripp et al. (2024, BUTTER-E) provide 63,527 watt-meter measurements across fully connected networks of varying size and shape, finding energy hardware- and cache-mediated rather than FLOPs-linear — evaluated on architecture design space, not post-training compression, with no reliability axis. Pachón, Pedraza & Ballesteros (2025) evaluated 180 pruned models, confirming strong batch-size dependence in the FLOPs–energy relationship. Deutel, Woller & Mutschler (2022) and Balaskas et al. (2023) measured physical power for compressed models on embedded hardware. Each reports accuracy as the sole quality metric alongside energy.

4.4 Software energy estimator validation

Fischer (2025) compared software estimators against external physical meters across hundreds of AI experiments and found errors up to 40%; this is a general-workload finding and is not presented here as specific to pyJoules or to compressed models. Aquino-Brítez et al. (2025) and Rodriguez et al. (2024) provide comparable general validation and review work. Kocher et al. (2025) found NVIDIA SMI can correlate poorly with an external power meter under some sampling conditions and proposed corrective calibration — directly relevant to the NVML behaviour independently observed in §2.4. de Paula, Soni, Upadhyay & Lagos (2025) is the closest prior work on the efficiency side, comparing pruning, quantization, and distillation for carbon efficiency via CodeCarbon, without hardware ground truth, a reliability axis, or a vision-domain application. Rojahn & Grum (2025), in a 103-study systematic review, conclude the field lacks reproducible, calibrated measurement across hardware tiers and calls for a hybrid estimator-plus-metering architecture — external corroboration of the gap this proposal addresses.

5. Research Gap

The hardware-energy literature and the reliability literature have each advanced substantially since 2019 without intersecting on a shared protocol. Studies that measure energy on compressed models report only accuracy; studies that measure calibration or class-level impact on compressed models report no energy.

Component combination	Representative prior work	Status
Compression + hardware energy	Pachón et al. 2025; Deutel et al. 2022; Balaskas et al. 2023; Tripp et al. 2024	Well covered — accuracy only
Compression + calibration	Mitra et al. 2024; Huber et al. 2025; Yuan et al. 2023	Covered — no energy; independently reconstructed here (§2.3)

Component combination	Representative prior work	Status
Compression + robustness	Shen et al. 2024; Diffenderfer et al. 2021; da Silva et al. 2025	Covered — no energy; context-dependent directions
Compression + class-level impact	Hooker et al. 2019, 2020; Tran et al. 2022; Paganini 2020	Mature — no energy
Energy + estimator validation	Fischer 2025; Aquino-Brítez et al. 2025; Kocher et al. 2025	General workloads — not tested on compressed models
Compression + measured energy + classwise calibration, one protocol	—	Not identified in our search; addressed here

Gap 1. Software estimators have not been tested on compressed models, where quantization changes the instruction mix on which they were calibrated. Framed here as an accounting-consistency audit (§3.1), not independent physical validation, which is reserved for Phase 2.

Gap 2. Reported reliability effects of pruning are context-dependent across studies, and a specific protocol confound in the closest study — unmatched recovery training — has not been tested. Preliminary evidence (§2.3) motivates a confirmatory test of this specific hypothesis.

Gap 3. Compression methods have not, to the extent our searches were able to establish, been compared at matched measured-energy budgets using statistical equivalence testing; existing comparisons normalise on FLOPs or nominal compression ratio — proxies whose validity is itself in question, and whose disconnect from real energy behaviour is demonstrated directly in §2.2.

6. Research Questions and Hypotheses

RQ1 — Instrument agreement. Do software energy estimators agree with the hardware counters they partly derive from, across the confirmed-executable compression grid (§2.2), and does that agreement change systematically as quantization precision decreases from FP32 to INT8 (ResNet-18, CPU)?

RQ2 — Recovery-training confound (Track A). Does the classwise calibration difference observed between a dense baseline and a globally-pruned, recovery-trained model (§2.3) persist when the baseline is given the same total training budget, or is the observed direction attributable to asymmetric training rather than to pruning?

RQ3 — Matched-energy selection risk (Track B). Among real structurally-pruned and quantized configurations of ResNet-18 and MobileNetV3-Small demonstrated to have statistically equivalent measured inference energy, does selecting the most accurate

configuration more often yield materially worse classwise calibration under corruption than an alternative eligible configuration?

RQ4 (Phase 2) — Physical and cross-platform generality. Do the Phase 1 findings on RQ1–RQ3 hold against external physical power measurement and on one additional hardware platform?

One primary confirmatory hypothesis is pre-specified for each of Tracks A and B; RQ1 is treated as a supporting methods result. All other comparisons are exploratory and reported as such, with Holm correction where multiple exploratory tests are drawn from the same family.

- **H1 (supporting, methods).** Estimator-counter disagreement, where the estimator and counter share an underlying interface, increases as quantization precision decreases from FP32 to INT8 on the confirmed CPU path. This is reported descriptively; no claim of independent instrument validity is made from it.
- **H2 (primary, Track A).** A dense VGG-19-BN model trained for the same total epoch budget ($160 + 40 = 200$) as the pruned-and-recovered model shows classwise ECE degradation, relative to a 160-epoch-only dense baseline, that is not distinguishable (via TOST, pre-specified margin ± 0.010 absolute ECE) from the degradation observed in the pruned-and-recovered model. Rejection of this equivalence — i.e. the pruned model is measurably worse even after the training-time confound is controlled — would indicate the effect is attributable to pruning itself, not training time.
- **H3 (primary, Track B).** Within pairs of ResNet-18/MobileNetV3-Small configurations shown by TOST to be energy-equivalent (measured-energy ratio within 0.95–1.05) and accuracy-equivalent (TOST, ± 0.010 absolute), the higher-accuracy member of each pair shows worst-class ECE degradation exceeding 0.010 absolute, relative to the alternative eligible member, more often than the exploratory ranking (ordinary accuracy-based selection) would predict by chance.

7. Phase 1, Track A — The Recovery-Training Confound (Months 1–6)

This track directly extends the reconstruction already completed (§2.3). It is the fastest-completing, most directly evidenced component of Phase 1.

WP1 — Matched-training dense control

Architecture, data, and hyperparameters exactly as in §2.3 (VGG-19-BN, CIFAR-10, SGD momentum 0.9, weight decay $1e-4$, batch 64, LR 0.1). Three arms per seed: (a) dense baseline, 160 epochs; (b) dense model, $160 + 40 = 200$ epochs, no pruning — the missing control in the reconstructed protocol; (c) globally pruned 30% + 40-epoch recovery, as already run. Ten seeds, extending the three already completed. Full clean and CIFAR-10-C evaluation for every arm and seed, using the same harness validated in §2.3–2.5.

Estimator, stated deliberately. Equal-width ECE is unstable when conditioned on class, given typical per-class sample counts; the primary calibration endpoint is classwise one-

versus-rest ECE with ten equal-mass bins and deterministic tie handling, matching the equal-mass convention already used in §2.3. Brier score is retained as a secondary measure of probabilistic predictive quality — it is not treated as an isolated calibration measure, since it reflects discrimination as well as calibration.

Analysis. H2 is tested via TOST with the pre-specified ± 0.010 absolute ECE margin (§6). The worst-minus-mean classwise gap is reported as a secondary, exploratory quantity, distinguished explicitly from worst-class error alone — the gap can rise because other classes improve, not because the worst class worsens, and the two are not conflated.

Output: a standalone finding, publishable independent of Track B’s outcome — either the training-time confound explains Mitra et al.’s reported direction, or pruning has an effect beyond it. Both are informative results.

8. Phase 1, Track B — Matched-Energy Selection Risk (Months 4–12)

This track uses only the compression configurations confirmed genuinely executable and energy-relevant in §2.2: real structural (channel) pruning via torch-pruning, and FP16/INT8 quantization on their confirmed-executable paths. Unstructured weight pruning is excluded from this track, having been confirmed to produce no measured energy benefit regardless of nominal sparsity (§2.2).

Architecture	Confirmed-executable arms
ResNet-18	FP32 baseline (GPU); FP16 (GPU); INT8 (CPU, torchvision.models.quantization.resnet18); real channel pruning 30/50/70% (GPU, via torch-pruning)
MobileNetV3-Small	FP32 baseline (GPU); FP16 (GPU); real channel pruning 30/50/70% (GPU, via torch-pruning) — no INT8 arm (§2.2)

Eleven confirmed-executable configurations across both architectures. Ten seeds each, following the validated compute budget of ≈ 1.3 days sequential GPU time for training (§2.5); energy measurement follows the validated protocol — minimum 1-second NVML measurement windows (§2.4), 30 paired measurement windows per confirmatory comparison, RAPL logged concurrently on the CPU-only INT8 arm.

WP2 — Energy-equivalence classification

Configuration pairs are classified as energy-equivalent using TOST against a pre-specified 0.95–1.05 ratio interval on measured joules per image, computed from at least 30 paired measurement windows per configuration. Configurations are also classified as accuracy-equivalent via TOST at ± 0.010 absolute top-1 accuracy. Failure to reject a difference is never treated as evidence of equivalence; only a confidence interval falling entirely inside the pre-specified margin is.

WP3 — Selection-risk test (H3)

Among pairs classified as jointly energy- and accuracy-equivalent, the member with (marginally) higher accuracy is compared against its pair partner on worst-class ECE under corruption. H3 is tested via the proportion of such pairs, across seeds and severities, in which the higher-accuracy member shows worst-class ECE degradation exceeding the pre-specified 0.010 absolute threshold.

Robustness metric, matching §2.3. Mean-per-corruption accuracy (mPC), averaged per severity over all 15 standard CIFAR-10-C corruptions — not normalised mCE, for direct comparability with the Track A reconstruction and with Mitra et al.’s own reported metric.

Timing and exclusion rules. Fixed batch size 64; warm-up passes discarded; a deterministic idle–active–idle measurement window (10s settling either side); background load logged; invalid runs (OOM, throttling detected via the clock-ceiling diagnostic of §2.4, failed convergence) are retained in the raw record and excluded from confirmatory analysis by a rule fixed in advance, never post hoc.

Multiplicity. H2 and H3 are the two pre-specified primary tests in this proposal. All ranking or ordering analysis beyond the specific pair-comparison in H3 (e.g. a full cross-configuration Kendall correlation) is exploratory, reported with Holm correction, and not used to support the thesis’s central claim.

9. Timeline

Note on research activity alongside coursework. The plan below sets out the research programme itself. In taught Master’s programmes, coursework typically runs alongside — particularly in year one — so research activity proceeds at reduced intensity during teaching periods and intensifies during dedicated thesis time. The preliminary work reported in §2 was deliberately produced ahead of enrolment for this reason: it reduces what remains to be done once coursework begins.

Months	Activity	Dependency
1–2	Track A: 7 additional VGG-19-BN seeds (dense-control arm added); harness extended to Track B architectures	Already-validated harness
2–4	Track A: full clean + corruption evaluation, all arms, 10 seeds; H2 analysis	Track A training
4	Track A complete: first paper drafted (publishable regardless of H2 outcome)	—
4–6	Track B: 10-seed training, ResNet-18 + MobileNetV3-Small, all 11 confirmed-executable configurations	Validated 1.3-day compute budget
6–9	Track B: energy measurement (≥ 30 windows/configuration), TOST energy- and accuracy-equivalence classification	Track B training complete
9–11	Track B: H3 selection-risk analysis; RQ1 estimator-agreement audit across full grid	WP2 equivalence classification

Months	Activity	Dependency
12	Phase 1 complete: two papers (Track A, Track B), public benchmark repository, arXiv preprint	—
13–15	Buffer quarter: onboarding, ethics/procurement for chosen Phase 2 extension	Placement
15–22	Phase 2: the single extension selected with supervisor (§10)	Lab access, supervision
22–24	Thesis writing; submission	—

Compulsory deliverables. A completed thesis; the reproducible evaluation pipeline already partially built (§2); a documented comparison on the two committed architectures under both tracks. Manuscript submission from Phase 1 is an intended output; acceptance is not treated as a completion requirement.

Slippage plan. If Track A overruns, seed count for the dense-control arm is reduced to 5 with the reduction reported, not silently absorbed. If Track B overruns, the MobileNetV3-Small arm is retained in full and the ResNet-18 INT8 arm is reported as incomplete rather than dropped without disclosure. Track B’s core structural-pruning comparison is never cut — it is the thesis question.

10. Phase 2 — One Bounded Extension (Months 13–24)

Phase 1 is presented as a complete, publishable thesis on its own. Phase 2 is not a catalogue of possible directions; it is one extension, to be selected jointly with a supervisor based on Phase 1’s actual findings and the specific resources and expertise available in their group. Four candidate extensions are described below with the specific resource each requires; the applicant proposes to adopt exactly one.

Candidate: physical power ground truth

Requires: bench power meter, current probes, oscilloscope, and expert guidance reconciling measurement boundaries. RAPL and NVML are themselves modelled estimates; this extension would establish the floor beneath the accounting-consistency audit in §3.1 and §6 (RQ1), converting it into genuine independent physical validation.

Candidate: cross-platform generality

Requires: access to at least one additional accelerator class (e.g. a server GPU with genuine sparse-kernel support, or an embedded/edge device). Tests whether Track B’s conclusions are specific to this consumer GPU’s lack of sparsity acceleration (§2.2) or generalise.

Candidate: modality generalisation

Requires: compute beyond the 6GB ceiling used throughout Phase 1, and domain supervision. Tests whether the Phase 1 protocol, applied to a small ($\approx 0.5\text{B}$) language model, reproduces the same qualitative pattern. UniComp (von Rad, Cao & Geiger, 2026)

occupies adjacent LLM-compression territory; the differentiator remains classwise probability calibration and measured energy, neither of which that work reports.

Candidate: real subgroup fairness

Requires: dataset access, ethics approval, and institutional affiliation. Phase 1's per-class analysis is a legitimate but distinct question from fairness across real demographic groups, which needs data an unaffiliated individual cannot access.

11. Expected Contributions

Already produced (reported in §2, prior to enrolment)

- An empirically verified executable compression grid for consumer vision hardware, including two confirmed dead ends (INT4 for CNNs; GPU-side INT8 via the tested stack) and one confirmed non-result (native PyTorch pruning APIs give zero energy benefit regardless of nominal sparsity).
- Two independently discovered instrument-characterisation findings: the NVML short-window power artifact, and GPU clock-ceiling as the correct diagnostic for CUDA-context contention on shared hardware.
- A 3-seed reconstruction of the closest prior study, showing a directionally consistent (6 of 6 conditions, sign-test $p \approx 0.016$) but not yet confirmatory disagreement with its reported direction on calibration.

Committed (Phase 1, thesis deliverable)

- A confirmatory 10-seed test of whether the reported calibration effect in the closest prior study is attributable to pruning or to an uncontrolled training-time confound (Track A, H2) — informative in either direction.
- The first evaluation, to our knowledge, of compression-method selection risk at TOST-established matched measured energy, using confirmed genuinely-executable structural compression rather than nominal compression ratios (Track B, H3).
- A reproducible benchmark with frozen protocol, released per-example outputs, and full provenance, covering both tracks.

Aspirational (Phase 2, one selected extension)

- Physical validation, cross-platform generality, modality generalisation, or real-subgroup fairness — one, selected with supervisor input.

Target venues. TMLR or Sustainable Computing: Informatics and Systems for the measurement and Track A findings; NeurIPS Datasets & Benchmarks or MLSys for the full framework; SustainML / EfficientML workshops for early-stage results.

12. Feasibility and Applicant Preparation

Section 2 of this proposal is itself the primary feasibility evidence: every claim of executability, every timing figure, and the central directional finding motivating Track A are backed by output already produced, not projected.

Technical background. Python, PyTorch, scikit-learn, pandas, NumPy, OpenCV; the full measurement and training harness described in §2 was built, debugged, and validated independently, including diagnosing and fixing one packaging bug in the reconstruction scaffold and one measurement-configuration error in the energy harness's default settings.

Methodological background. The applicant's undergraduate thesis reconstructed three decades of land-cover change from satellite imagery using multiple classifiers under spatially-blocked rather than random validation. That single design choice reversed the headline conclusion, while the choice of classifier moved the result by roughly five percentage points. The full pipeline was released publicly, and the work is under peer review at ICCIT 2026. A related literature review, surveying the energy-measurement literature this proposal builds on, is separately under revision following peer review.

The lesson from that thesis — that evaluation design, not algorithm choice, determined the finding — is the direct origin both of this proposal and of the specific confound this proposal now proposes to test (§2.3, §6, H2). The subject matter has changed; the disposition has not.

What scholarship support enables. Existing hardware allows preliminary work to continue independently, as §2 demonstrates. Scholarship support provides sustained research time free of the applicant's current full-time employment, formal statistical and methodological training, and supervision to carry the confirmatory design in §6–§8 through to completion and, in Phase 2, to extend it into regimes (physical instrumentation, larger-scale compute, real demographic data) that are not reachable independently. The applicant intends to release the full pipeline and results publicly, usable on comparably modest hardware, including by researchers and students without access to institutional compute.

Stated limitations. No accepted publications to date; one manuscript under review, one under revision. Mathematical background is sufficient to apply and extend established statistical methods (TOST, paired testing, sign-test framing, as used throughout §2 and §6–§8) with reference to the literature, rather than to derive new theory independently.

13. Risks and Mitigation

Risk	Mitigation
H2 or H3 is not confirmed	Both are pre-registered as informative in either direction. A well-powered null result on H2 (training time fully explains the effect) or H3 (no selection risk at matched energy) is itself publishable, because the equivalence margins are fixed in advance rather than chosen post hoc.
Measurement noise exceeds the effect	The 1-second minimum NVML window and the 30-window-per-configuration requirement are both derived from measured

Risk	Mitigation
	instrument behaviour (§2.4–2.5), not assumed.
Prior work is closer than assessed	Positioning (§4) states specific extension axes rather than an empty field, and the closest work (Mitra et al.) has been fully reconstructed and run, not merely cited.
Compute or timeline overrun	The full Track A + Track B training budget (≈ 2.9 days sequential across both tracks) is a direct hardware measurement, not an estimate; the slippage plan (§9) specifies exactly what is cut first.
Concurrent work occupies the space	Track A's first result depends only on already-completed infrastructure and is the fastest output (month 4); early preprinting establishes priority.

14. Why Supervision Is Sought

This proposal is not a request to begin a project; the preliminary work in §2 demonstrates a project already under way, with a specific, evidenced, falsifiable confirmatory question. What that work cannot do alone is precisely what supervision and a funded position provide: sustained time free of full-time employment, formal training in the statistical methods this design already uses at an applied level, and — for Phase 2 specifically — access to physical instrumentation, additional hardware platforms, or demographic data that cannot be acquired independently.

The specific request: supervision of a Master's thesis testing whether a reported compression-calibration relationship survives a training-time confound (Track A), and whether compression-method selection at matched measured energy carries a reliability risk not visible under standard accuracy-only comparison (Track B) — with the direction of the final year's single extension (§10) to be agreed jointly once Phase 1's findings are in.

References

- Aquino-Brítez, A., García-Sánchez, P., & Ortíz, A. (2025). Towards an energy consumption index for deep learning models: a comparative analysis of architectures, GPUs, and measurement tools. *Sensors*, 25(3), 846. DOI: 10.3390/s25030846
- Balaskas, K., Karatzas, A., Sad, C., Siozios, K., & Anagnostopoulos, I. (2023). Hardware-aware DNN compression via diverse pruning and mixed-precision quantization. *IEEE Transactions on Emerging Topics in Computing*. DOI: 10.1109/TETC.2023.3346944
- Blakeney, C., Huish, N., Yan, Y., & Zong, Z. (2021). Simon says: evaluating and mitigating bias in pruned neural networks with knowledge distillation. *arXiv:2106.07849*
- da Silva, I. W. D., Pereira, E., Barboza, E. de A., dos S. Neto, B. F., & Ribeiro, M. de M. (2025). Evaluating the impact of compression techniques on the robustness of CNNs under natural corruptions. *ICMLA 2025*. DOI: 10.1109/ICMLA66185.2025.00055

- de Paula, E., Soni, J., Upadhyay, H., & Lagos, L. (2025). Comparative analysis of model compression techniques for achieving carbon efficient AI. *Scientific Reports*. DOI: 10.1038/s41598-025-07821-w
- Deutel, M., Woller, P., & Mutschler, C. (2022). Deployment of energy-efficient deep learning models on Cortex-M based microcontrollers using deep compression. *arXiv:2205.10369*
- Diffenderfer, J., Bartoldson, B., Chaganti, S., et al. (2021). A winning hand: compressing deep networks can improve out-of-distribution robustness. *arXiv:2106.09129*
- Fischer, R. (2025). Ground-truthing AI energy consumption: validating CodeCarbon against external measurements. *it – Information Technology*. DOI: 10.1515/itit-2025-0031; *arXiv:2509.22092*
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. *ICML*.
- Hendrycks, D., & Dietterich, T. (2019). Benchmarking neural network robustness to common corruptions and perturbations. *ICLR*. *arXiv:1903.12261*
- Hong, J., Duan, J., Zhang, C., et al. (2024). Decoding compressed trust: scrutinizing the trustworthiness of efficient LLMs under compression. *ICML, PMLR 235*. *arXiv:2403.15447*
- Hooker, S., Courville, A., Clark, G., Dauphin, Y., & Frome, A. (2019). What do compressed deep neural networks forget? *arXiv:1911.05248*
- Hooker, S., Moorosi, N., Clark, G., Bengio, S., & Denton, E. (2020). Characterising bias in compressed models. *arXiv:2010.03058*
- Huber, P., Göhner, U., & Trapp, M. (2025). Comprehensive analysis of neural network inference on embedded systems: response time, calibration, and model optimisation. *Sensors*, 25(15), 4769. DOI: 10.3390/s25154769
- Kamal, M., & Talbert, D. (2025). Downsized and compromised? Assessing the faithfulness of model compression. *arXiv:2510.06125*
- Kocher, M., et al. (2025). Guidelines for the quality assessment of energy-aware NAS benchmarks. Leibniz Universität Hannover.
- Mitra, P., Schwalbe, G., & Klein, N. (2024). Investigating calibration and corruption robustness of post-hoc pruned perception CNNs: an image classification benchmark study. *CVPR Workshops*, 3542–3552. DOI: 10.1109/CVPRW63382.2024.00358; *arXiv:2405.20876*
- Paganini, M. (2020). Prune responsibly. *arXiv preprint*.
- Pachón, C. G., Pedraza, C., & Ballesteros, D. (2025). PruneEnergyAnalyzer: an open-source tool for measuring energy consumption in pruned neural networks. *Big Data and Cognitive Computing*, 9(8), 200. DOI: 10.3390/bdcc9080200
- Rodriguez, C., Degioanni, L., Kameni, L., Vidal, R., & Neglia, G. (2024). Evaluating the energy consumption of machine learning: systematic literature review and experiments. *arXiv:2408.15128*
- Rojahn, M., & Grum, M. (2025). Green AI: a systematic review and meta-analysis of its definitions, lifecycle models, hardware and measurement attempts. *arXiv:2511.07090*
- Shen, L., Edalati, A., Meyer, B. H., Gross, W. J., & Clark, J. J. (2024). Robustness to distribution shifts of compressed networks for edge devices. *arXiv:2401.12014*

- Tran, C., Fioretto, F., & Kim, J. (2022). Pruning has a disparate impact on model accuracy. arXiv:2205.13574
- Tripp, C., Perr-Sauer, J., Gafur, J., Nag, A., Purkayastha, A., Zisman, S., & Bensen, E. (2024). Measuring the energy consumption and efficiency of deep neural networks: an empirical analysis and design recommendations (BUTTER-E). arXiv:2403.08151; dataset DOI: 10.25984/2329316
- von Rad, J., Cao, Y., & Geiger, A. (2026). UniComp: a unified evaluation of large language model compression via pruning, quantization and distillation. EMNLP 2026. arXiv:2602.09130
- Williams, M., & Aletras, N. (2023). How does calibration data affect the post-training pruning and quantization of large language models? arXiv:2311.09755
- Yuan, Z., Liu, J., Wu, J., Yang, D., Wu, Q., Sun, G., et al. (2023). Benchmarking the reliability of post-training quantization: a particular focus on worst-case performance. arXiv:2303.13003